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Strong is the New Skinny, but is it Ideal?: A Test of the Tripartite Influence Model using a new Measure of Fit-Ideal Internalisation

Abstract

Although the thin body ideal has been the epitome of western female beauty for decades, the more recent fit body ideal is becoming arguably more popular. This study aimed to test two versions of the tripartite influence model (TIM; Thompson et al., 1999): one that included thin-ideal internalisation, and one that included fit-ideal internalisation, using path analysis. Participants were 558 Australian women aged 16-50 years (M = 22.06, SD = 7.41). All of the hypothesised paths were significant for both the thin- and fit-ideal internalisation models. The final models for both thin- and fit-ideal internalisation included extra, non-hypothesised paths from media to body dissatisfaction, and from social comparison to dieting, bulimic symptoms, and dieting. The two models differed in that the final model for thin-ideal internalisation included direct paths from thin-ideal internalisation to dieting and bulimic symptoms. The final model for fit-ideal internalisation, however, included a direct path from fit-ideal internalisation to compulsive exercise. It was concluded that, in contrast to the widely held perception that the fit ideal is a 'healthier' alternative to the thin ideal, fit ideal internalisation may be detrimental to female body satisfaction, disordered eating, and compulsive exercise.

KEYWORDS: fit ideal, thin ideal, body ideal, internalization, tripartite influence model, body image

1. Introduction

'Beauty' is a socially and culturally fluid construct. In western society, there is considerable emphasis placed on body shape and size in the determination of whether or not a woman is considered beautiful (Smith et al., 2007; Swami & Tovée, 2007). However, the exact composition of an 'ideal' societal body standard has changed considerably over time (Anderson, Crawford, Nadeau, & Lindberg, 1992). During times of financial hardship when food was scarce, such as the Renaissance period for example, women were considered beautiful if they were well rounded with significant body fat, as such a body type represented wealth and prosperity (Anderson et al., 1992; Calogero, Boroughs, & Thompson, 2007). In times of relative prosperity, and for decades now, a thin female body is viewed in western culture as desirable and beautiful (Swami et al., 2010). Indeed, embodied in Kate Moss' famous words "nothing tastes better than skinny feels", western culture not only holds a thin body ideal up as the epitome of female beauty, but also ascribes positive personality and social attributes (such as social attention, romantic success, employment and economic success) to those with a thinner body (Engeln-Maddox, 2006; Grogan, 1999).

Unfortunately, the 'thin ideal' is vastly different from the body of the average woman and is very difficult for the average woman to attain, leading many to experience body image disturbance and to engage in unhealthy eating behaviours (Cash, 2011; Moradi, Dirks & Matteson, 2005; Stice, 2001; Striegel-Moore & Bulik, 2007; Thompson, Heinberg, Altabe, & Tantleff-Dunn, 1999). In order to explain the antecedents and consequences of thin idealisation, Thompson et al. (1999) put forward the tripartite influence model (TIM; see Figure 1). The TIM suggests that societal pressures to be thin come from three sources: family, peers, and the media. Such pressures lead women to compare themselves to others (social comparison) and to 'buy into' and accept the notion that to be beautiful one must be thin (thin-ideal internalisation). According to the TIM, both social comparison and thin-ideal

internalisation lead women to feel dissatisfied with their bodies, which in turn leads them to engage in unhealthy eating behaviours directed towards attaining what they perceive to be the ideal thin body. There has been considerable support for the relationships between constructs incorporated within the TIM, as well as for the model in its entirety (e.g., Blowers, Loxton, Grady-Flesser, Occhipinti, & Dawe, 2003; Keery, van den Berg, & Thompson, 2004; Rodgers, Chabrol, & Paxton, 2011; Shroff & Thompson, 2006; van den Berg, Thompson, Obremski-Brandon, & Coovert, 2002). Thus, the thin ideal and women's relentless pursuit of it, have repeatedly been shown to be detrimental to women both physically and psychologically.

Although the thin ideal has been the dominant standard of female beauty for decades within western society, there is now a competing body ideal that is becoming arguably more prevalent. The 'fit' ideal, a body ideal that is simultaneously lean and toned, has gained increasing popularity, particularly within the last decade. Indeed, social media content perpetuating messages that 'strong is the new skinny' are commonplace (Boepple, Ata, Rum, & Thompson, 2016), and underscore the popularity of the female fit ideal. It has been found that more women strive to achieve a simultaneously lean and toned body, than to achieve a muscular or thin body alone (Kelley, Neufeld, & Musher-Eizenman, 2010). Similarly, when asked to choose their ideal body shape, it has been found that more women choose a model who is both lean and toned, rather than just thin (Bozsik, Whisenhunt, Hudson, Bennett, & Lundgren, 2018; Uhlmann, Donovan & Zimmer-Gembeck, under review).

Social media platforms such as Instagram may well have been the genesis of the fit body ideal, and imagery on these platforms serves to constantly bombard women with fit body images, quotes, and advice about fitness and nutrition (Boepple et al., 2016; Tiggemann & Zaccardo, 2016). Social media is one of the most common sources of health and fitness information for young women (Jong & Drummond, 2016), with over 50% believing it to be a

reliable and valid source of information (Raggatt et al., 2018). In fact, there are over 100 million hashtags or references (e.g., fitspo, fitspiration) on Instagram alone that relate to the fit ideal (Instagram, 2020). In addition to constant social media endorsement of the fit ideal, booming fitness and supplement industries promote it as being 'healthy', with advertisements claiming to assist women to achieve a fit body through exercise and supplement use. Indeed, there appears to be somewhat of a backlash against the thin ideal, with women rating 'fit' female avatars higher than thin female avatars in terms of discipline, life control, self-confidence, mental health and physical health (Uhlmann et al., under review). Moreover, women rate thin female avatars lower than overweight female avatars in terms of mental health and self-confidence, and rate thin and overweight female avatars similarly in terms of life-control and physical health (Uhlmann et al., under review). Thus, there is some evidence to suggest that the fit ideal may be slowly replacing the thin ideal as the most popular western female body standard.

The overwhelming physical and mental health benefits of exercise are indisputable (Hausenblas & Fallon, 2006; Martin Ginis, Strong, Arent, Bray, & Bassett-Gunter, 2014), and therefore it may seem logical to view a change towards a 'fit' sociocultural body ideal, as a positive one. However, no body ideal is ideal. In western societies, where a woman's body is considered an integral part of her value (Stice, 1994), the existence of any 'ideal' means that women will try to emulate it. Fitness models are every bit as 'lean' (in terms of body fat) as thin models, and they also have significant musculature (Tiggemann & Zaccardo, 2016). Such a body is extremely difficult for the average woman to achieve, and likely requires significant dietary changes, supplement use, and exercise (Uhlmann, Donovan, & Zimmer-Gembeck, 2019). Thus, the fit body ideal is arguably more difficult to attain than the thin body ideal. In addition, it has been found that although 90% of women who access 'fitspo' related social media sites report doing so for health-related reasons, 54% turn to these sites

for appearance-related reasons (Raggatt et al., 2018). Furthermore, 73% of women who desire muscle tone engage in exercise behaviour to change their body shape, and 26% change their diet and take supplements for the same reason (Kyrejto, Mosewich, Kowalski, Mack, & Crocker, 2008). Thus, many women are not only engaging in exercise to be healthy, they are also exercising in order to enhance their appearance – and from a sociocultural perspective of body image disturbance, that is where the problem lies (Tiggemann, 2011).

Despite representation and promotion by the fitness industry, supplement industry, and social media that the fit ideal is healthy, there is evidence to suggest that this is not in fact the case. For instance, between 30-57% of women report feeling dissatisfied with their muscle tone (Fallon, Harris, & Johnson, 2014; Fiske, Fallon, Blissmer, & Redding, 2014) and 18-21% of college women report engaging in compulsive exercise behaviour (Guidi et al., 2009). In addition, exposure to fit ideal images has been found to result in greater body dissatisfaction than exposure to thin ideal images (Robinson et al., 2017), and has been shown to lead to negative mood (Prichard, McLachlan, Lavis, & Tiggemann, 2017). Furthermore, it has been found that those who engage with fitspiration media report high levels of compulsive exercise, drive for thinness, drive for muscularity and bulimic symptoms, with 17.7% being at high risk of an eating disorder, 17.4% exhibiting very high levels of psychological distress, and 10.3% reporting addictive exercise behaviour (Holland & Tiggemann, 2017; Raggatt et al., 2018). Alarmingly, 1.6% of young girls aged 12-18 years report using protein shakes, creatine, amino acids, growth hormones and steroids at least weekly in order to increase their muscle tone (Field, 2005). Thus, it would seem that 'fit' does not necessarily equate to 'health' when it comes to body ideals.

Although the above discussion paints a relatively negative picture of outcomes associated with exposure to, or engagement with, fit ideal media, attempts to examine the antecedents and consequences of internalising the fit ideal (as proposed by the TIM) have

been hindered to date by the absence of a specific fit-ideal internalisation measure. In the small amount of research conducted thus far, researchers have used the 'athletic' internalisation subscale of the Sociocultural Attitudes Towards Appearance Questionnaire (SATAQ-3, SATAQ-4; Calogero, Davis, & Thompson, 2004; Schaefer et al., 2015) to measure fit-ideal internalisation. The resulting research has produced mixed results, and on balance suggests that 'athletic' ideal internalisation is less detrimental than the thin ideal. For instance, Ramme, Donovan, and Bell (2016) found no evidence that athletic internalisation was associated with body dissatisfaction or acted as a mediator in the relationship between sociocultural influences and body dissatisfaction. Similarly, Bell, Donovan, and Ramme (2016) found no association between athletic internalisation and body dissatisfaction, but found a direct association between athletic ideal internalisation and dieting, bulimic behaviours and compulsive exercise. Hoffmann and Warschburger (2019) found that athletic ideal internalisation predicted muscularity concerns and muscularity-oriented behaviours in female adolescents and young adults. Finally, Homan (2010) found that athletic ideal internalisation predicted compulsive exercise at a seven month follow up, but did not predict dieting or body dissatisfaction.

The problem with the research conducted to date in this area, is that the athletic ideal subscale of the SATAQ measures 'muscular' rather than 'fit' ideal internalisation, and in fact has been labelled as such in the most recent version of the assessment tool (SATAQ-4R; Schaefer, Harriger, Heinberg, Soderberg, & Thompson, 2017). That is, items within this subscale relate only to musculature, just as items within the thin ideal internalisation subscale of the SATAQ, relate only to thinness. As noted above however, the fit ideal is an amalgamation of the two; a body that is simultaneously both lean *and* toned. In an attempt to circumvent this problem, Uhlmann, Donovan, Zimmer-Gembeck, Bell, and Ramme (2018) examined whether concurrent muscular internalisation would mitigate the relationship

between thin internalisation and a range of negative outcomes, in line with societal promotion that the fit ideal is a 'healthy' ideal to which women should aspire. The authors found that incorporating muscularity into an ideal of thinness did not reduce the thin ideal's effects on negative affect, body dissatisfaction, bulimic symptoms or dieting, thus suggesting that aspirations for a fit body are no less detrimental than aspirations for a thin body. The authors noted that using moderation analyses to combine thin and muscular internalisation however, provided only a proxy measure of fit internalisation, as the ideal is likely to be greater than the mere sum of its parts. The Fit-Ideal Internalisation Test (FIIT; Uhlmann et al., 2019) was therefore developed to specifically assess fit-ideal internalisation.

In the construction of the FIIT, definitions of internalisation across various literature were collated, and internalisation was defined as a multidimensional construct that encompasses three domains: (1) cognitive affective acceptance of, and desire to conform to, a societal standard (i.e., idealisation), (2) cognitive and affective integration of the value and meaning of that standard into a personal value system (i.e., overvaluation), and (3) compliance with, or engagement in, value/norm congruent behaviour (i.e., behavioural drive). In the psychometric evaluation of the scale, a three-factor structure of fit-ideal internalisation was confirmed, and strong support was found for its internal consistency and test-retest reliability. Supporting the utility of a total score, good model fit was found for a higher-order factor, and the new scale was found to be associated with objectified body consciousness, drive for muscularity, drive for thinness, perfectionism, exercise for appearance reasons, and body comparison, but not exercise for health reasons. Furthermore, demonstrating that the fit ideal constitutes a distinct body ideal from that of thinness and / or muscularity alone, fit-ideal internalisation was significantly associated with bulimic symptoms, dieting, psychological distress, body dissatisfaction and compulsive exercise, above and beyond the effects of both thin-ideal and muscular-ideal internalisation, as measured by the SATAQ-4R.

Thus, measuring fit-ideal internalisation directly appears to provide more nuanced information than combining existing thin- and muscular-ideal internalisation scales (Uhlmann et al., 2019).

1.1 Aims and Hypotheses

The TIM was developed and tested in relation to the thin ideal, and before the advent of social media and the subsequent rise of the fit ideal. The aim of this study was therefore to test two versions of the TIM: one that included the original thin-ideal internalisation construct (Figure 2), and one that included the more recent fit-ideal internalisation construct (Figure 3), in order to assess whether the TIM is useful in explaining the antecedents and consequences of fit-ideal internalisation. It was decided to test two separate models rather than to include fit- and thin-ideal internalisation within the same model, as the primary research question was to test whether or not the TIM applies to fit-ideal internalisation, not to test a 'new' model that might incorporate the two ideals. The use of thin-ideal internalisation in this study was therefore for comparison purposes only.

As is evident from Figures 2 and 3, there were two minor differences between the proposed models and the original TIM. First, a path from thin/fit-ideal internalisation to social comparison (rather than the other way around) was included for theoretical reasons and in line with the empirical findings of others (e.g., Fitzsimmons-Craft, Harney, et al., 2012). Second, 'compulsive exercise' was included within the model as a potential outcome variable due its relevance to the fit ideal. It was hypothesised that for both models, there would be significant paths from family, peer and media pressures to thin/fit-ideal internalisation and social comparison; that there would be significant paths from thin/fit-ideal internalisation and social comparison to body dissatisfaction; and that there would be significant paths from body dissatisfaction to dieting, bulimic behaviours and compulsive exercise. It was also

hypothesised that there would be a significant path from thin/fit ideal internalisation to social comparison.

2. Method

2.1 Participants

Table 1 presents participant demographic information. Participants were 558 Australian women aged 16-50 years (*M* = 22.06, *SD* =7.41), with BMIs ranging from 16 to 49.34 (*M* = 23.57, *SD* = 5.20). The majority of women were born in Australia (76.3%), reported that high school was their highest attained level of education (61.8%), and had a household income below $AUD 90,000 (69.5%). The demographic features of the sample (i.e., a young Caucasian undergraduate female sample) are consistent with the samples described in previous studies in this area (e.g., Tiggermann & Zaccardo, 2015) and are consistent with the demographic of women shown to engage with social media content (Holland & Tiggemann, 2016).

2.2 Measures

2.2.1. Sociocultural Pressures

Peer, family, and social media pressures were measured using modified versions of the Pressures Peer, Pressures Family, and Pressures Media subscales of the Sociocultural Attitudes Towards Appearance Questionnaire – Version 4 Revised (SATAQ-4R; Schaefer et al., 2017). The original subscales consist of 4 items measuring peer, family, and traditional media pressures to conform to body ideals. One item assessing pressure to be toned was also added to each subscale (i.e., “I feel pressure from my family to look more toned”, “I feel pressure from my peers to look more toned”, “I feel pressure from media to look more toned). The Media subscale was also extended to include 5 additional items where the words ‘media’ were replaced by ‘social media’ so that both traditional media (newspapers, magazines, TV, etc.) and social media were included. The Family and Peer subscales, therefore, consisted of

5 items, and the Media subscale consisted of 10 items. Each item required respondents to rate their agreement with each item on a 5-point scale from 1 (*definitely disagree*) to 5 (*definitely agree*). Scores on each subscale were summed to produce total subscale scores that could range from 5 to 25 for the Family and Peers subscales, and from 10-50 for the Media subscale, with higher scores indicating greater pressure from peers, family, and media. Cronbach's alphas of .93, .92, and .96 have been demonstrated for the original Pressures Family, Pressures Peer, and Pressures Media subscales, respectively (Schaefer et al., 2017). In this study, Cronbach's alphas for the modified subscales were found to be .93, .95, and .97 for the Pressures Family, Pressures Peer, and the extended Pressures Media subscales, respectively.

2.2.2. Thin-ideal internalisation

Thin-ideal internalisation was measured using the 4-item Thin-Ideal Internalisation subscale of the SATAQ-4R described above (Schaefer et al., 2017). Respondents are required to rate on a 5 point scale from 1 (*definitely disagree*) to 5 (*definitely agree*) the extent to which they agree with each item (e.g., "I want my body to look very thin"). Scores are summed to produce a total score that may range from 5 to 20, with higher scores indicating greater thin-ideal internalisation. The Thin-Ideal Internalisation subscale has demonstrated acceptable internal consistency in previous studies (α= .82; Schaefer et al., 2017), and a Cronbach's coefficient alpha in the current study of .90.

2.2.3. Fit-ideal internalisation

Fit-ideal internalisation was measured using the total score of the Fit-Ideal Internalisation Test (FIIT; Uhlmann et al., 2019). The 20-item FIIT consists of three subscales that measure fit ideal internalisation (8 items, e.g., "I am preoccupied with the idea of having a body that looks both lean and toned"), overvaluation (8 items; e.g., "Having a body that is both very lean and very toned is a good way to show people you are in control of

your life"), and behavioural drive (4 items; e.g., "I spend time doing things to ensure my body looks both lean and toned"). Respondents rate their agreement with each item on a 5-point scale from 1 (*strongly disagree*) to 5 (*strongly agree*). For the purposes of the present study, the total score was used to provide the most comprehensive measure of fit-ideal internalisation. All items are summed to produce the total score that may range from 5 to 100, with higher scores indicating higher levels of fit-ideal internalisation. Providing further support for use of the total score, the FIIT composite score has shown excellent internal consistency, with a Cronbach's alpha of .94, as well as a test-retest reliability of .77 over a two-week period (Uhlmann et al., 2019). The FIIT has also demonstrated good convergent, discriminant, and incremental validity: predicting disordered body image, eating and exercise outcomes, above and beyond that of thin-ideal and muscular-ideal internalisation (Uhlmann et al., 2019). The Cronbach's alpha of the FIIT total score in the current study was .94.

2.2.4 Social Comparison

The 18-item Body, Eating and Exercise Orientation Comparison Measure (BEECOM; Fitzsimmons-Craft, Bardone-Cone, & Harney, 2012) was used to measure social comparison. Respondents are required to rate the frequency of each item (e.g., "I compare my body shape to that of my peers") on a 7-point scale from 1 (*never*) to 7 (*always*). Scores on each item are summed to produce a score that may range from 18 to 126, with higher scores indicating a greater propensity to engage in social comparison. The BEECOM total score has shown good internal consistency, with a Cronbach's alpha of .97, as well as good validity (Fitzsimmons-Craft, et al., 2012). The Cronbach's alpha of the BEECOM in the present study was .98.

2.2.5 Body Dissatisfaction

The Body-Image Ideals Questionnaire (BIQ; Cash & Szymanski, 1995) was used to measure body dissatisfaction. This measure was chosen over other more commonly used measures due to its greater relevance to the fit ideal. The most popular measures of body

dissatisfaction operationalise the construct in terms of level of dissatisfaction with one's weight or body fat. The BIQ does not assume thinness as the only source of body dissatisfaction and was therefore chosen for its greater applicability to the fit ideal. Respondents are required to rate: (a) how close each of 11 attributes (e.g., muscle tone, weight, height) are to their personal ideal (i.e. discrepancy) on a 4-point scale from 0 (*exactly as I am*) to 3 (*very unlike me*), and (b) how important each physical ideal is to them on a 4-point scale from 0 (*not important*) to 3 (*very important*). Items on the discrepancy scale are recoded such that 0 is scored as -1. For each item, the cross product of the discrepancy and importance scales is computed. The cross-products for each item are then summed and averaged to provide a total score that may range from -3 to +9, with higher scores indicating greater body dissatisfaction. The BIQ has shown acceptable reliability, with a Cronbach's alpha of .76 (Cash & Szymanski, 1995). The Cronbach's alpha of the BIQ in the current study was .83.

2.2.6 Dieting & Bulimic Symptoms

Dieting and bulimic symptoms were measured using the 13-item Dieting subscale and the 8-item Bulimia and Food Preoccupation subscales respectively of the Eating Attitudes Test (EAT-26: Garner, Olmsted, Bohr, & Garfinkel, 1982). Respondents are required to rate on a 7-point scale from 0 (*never*) to 6 (*always*), the extent to which each item (e.g., Dieting: "I am aware of the calorie content of foods that I eat"; Bulimia and Food Preoccupation: "I vomit after I have eaten") applies to them. Items are recoded such that scores below 3 are recoded as 0, and scores from 4 to 6 are recoded as 1, 2 and 3, respectively. Scores for each subscale are then summed to produce a total score that may range from 0 to 39 for the Dieting subscale, and from 0 to 24 for the Bulimia and Food Preoccupation subscale, with higher scores indicating higher dieting and bulimic behaviours respectively. The Dieting and Bulimia and Food Preoccupation subscales have shown good reliability, with Cronbach's

alphas of .90 and .84 respectively (Garner et al., 1982). The Cronbach's alphas of the Dieting and Bulimia and Food Preoccupation subscales in the current study were .90 and .83, respectively.

2.2.7 Compulsive Exercise

Compulsive exercise behaviour was measured using the 24-item Compulsive Exercise Test (CET; Taranis, Touyz, & Meyer, 2011). Respondents are required to rate how true each item is for them on a 6-point scale from 0 (*never true*) to 5 (*always true*). The total score is derived by summing the averages of each of the five subscales: (1) avoidance and rule driven behaviour – 8 items (e.g., "If I cannot exercise I feel low or depressed"), (2) weight control exercise – 5 items (e.g., "I exercise to improve my appearance"), (3) mood improvement – 5 items (e.g., "I feel less anxious after I exercise"), (4) lack of exercise enjoyment – 3 items (e.g., "I do not enjoy exercising"), and (5) exercise rigidity – 3 items (e.g., "My weekly pattern of exercise is repetitive"). The total score may therefore range from 0 to 25, with higher scores indicative of higher compulsive exercise behaviours. The CET has been found to have acceptable internal consistency, with a Cronbach's alpha of .83 (Taranis et al., 2011). The Cronbach's alpha of the CET in the current study was .90.

2.3 Procedure

Ethical approval was sought and granted through the Griffith University Human Ethics Committee. Participants were recruited through the first year Psychology student pool, the University call for research volunteers, and social media (Facebook). Interested individuals were provided with a link directing them to the downloadable online information and consent forms. After consent was obtained, participants were directed to a Lime Survey site where they completed the online battery of questionnaires described above. Participation was anonymous. Participants were provided the opportunity to enter their names into a draw to win a $100 Amazon gift voucher. Students were asked to provide their names to gain course

credit. Names and questionnaires were kept separate to ensure anonymity and confidentiality of responses.

3. Results

3.1 Statistical Analyses

Of the 636 women who signed up for the study, 73 failed to go on to complete any of the questionnaires in the battery. Data for the remaining 563 women were checked for missing data, skew, kurtosis, and multivariate outliers. There were no other missing data. Moderate levels of skew were evident on social pressure variables, body dissatisfaction, dieting, and bulimic symptoms as expected. Bootstrapping procedures were used to account for skewed data and to obtain bias-corrected confidence intervals for the indirect effects. Bollen–Stine bootstrap was calculated for the chi-square test to correct for inflated standard errors due to skew (Bollen & Stine, 1992). In addition, Mahalanobis Distance identified five multivariate outliers that were subsequently removed from the structural equation modelling analysis, leaving the final sample of 558 participants.

Path analysis was used to test the proposed model using AMOS 25 statistical software with maximum likelihood estimation using the covariance matrix (Arbuckle, 2017). The model fit was evaluated with chi-square (χ^2) goodness of fit index statistic. For a fit to be acceptable, chi-square should be non-significant. However, the significance of χ^2 is influenced by sample, such that sample sizes as large as that used in our study will result in a significant χ^2, even if there are minor differences between the observed data and the hypothesised model. As such, we also used the following indices of fit: normed χ^2 (χ^2 /df), Tucker-Lewis index (TLI), comparative fit index (CFI), root mean squared error of approximation (RMSEA), and standardised root mean square residual (SRMR). For acceptable fit, CFI and TLI should be > .90, RMSEA should be < .06, and SRMR should be < .08 (Hu & Bentler, 1999). The normed χ^2 should be between 1 and 3 (Kline, 2004).

3.2 Structural Path Models

Table 2 shows the intercorrelations and descriptive statistics for the variables included in the tested models. As is evident from Table 2, all variables in the model were significantly correlated as expected. There was a strong correlation between thin-ideal and the fit-ideal internalisation variables. Two hypothesised models were tested. The two models were identical apart from the type of body ideal internalised. The first model tested thin-ideal internalisation and the second tested fit-ideal internalisation. Figures 2 and 3 show the models tested. Given the moderately strong correlations between compulsive exercise, dieting, and bulimic symptoms, error variances between these three variables were allowed to covary in both models.

3.2.1 Thin-ideal internalisation. As shown in Table 3, the χ^2 was significant and the other indices showed that the hypothesised model (Model 1) had poor fit with the data. Modification indices were assessed for potential additional paths that could improve the overall fit of the hypothesised model (i.e., show a significant decrease in χ^2 relative to the loss of degrees of freedom). A nested models approach was used, whereby additional paths that were suggested by the modification indices and theoretically plausible were freed (i.e., a direct path is added between two variables) in a sequential manner until the model achieved adequate fit. This approach minimises the possibility of including paths that improve fit while also achieving parsimony. Model 2 included a path between social comparison and compulsive exercise. As shown in Table 3, this was a significant improvement (evidenced by a significant change in χ^2). However, the overall model continued to demonstrate poor fit across all indices, and modification indices were assessed for potential paths to free. This same approach was used until there were no further theoretically justified paths to free. Model 3 included a free path from body dissatisfaction to dieting, Model 4 included a free path from media pressure to body dissatisfaction, Model 5 included a free path from thin-

ideal internalisation to bulimia, and Model 6 included a free path from social comparison to dieting. Again, while each freed path significantly improved fit, the RMSEA continued to demonstrate a less than optimal fit at Model 6. A final suggested path from social comparison to bulimia was freed in the final model (Model 7). The normed χ^2 was within the desirable range of between 1 and 3 (Kline, 2004). No further paths were suggested to be freed. All paths in the final model were significant at the $p < .05$. Although the χ^2 for the model was significant (Bollen Stine correction, $p = .003$), all other fit indices (including the normed χ^2) showed this to be a well-fitting model (see Figure 4).

3.2.2 Fit-ideal internalisation. As shown in Table 4, the χ^2 was significant and the other indices showed that the hypothesised model (Model 1) had poor fit with the data. Other fit indices also indicated poor fit of the model to the data. Again, modification indices were assessed to suggest potential paths that could improve the overall fit of the hypothesised model. Model 2 included a path from fit-ideal internalisation to compulsive exercise. As shown in Table 4, although this was a significant improvement, the overall model continued to demonstrate poor fit across all indices. Model 3 included a free path from social comparison to dieting, Model 4 included a free path from social comparison to bulimic symptoms, and Model 5 included a free path from social comparison to compulsive exercise. Again, although each freed path significantly improved fit, the RMSEA and the normed χ^2 continued to demonstrate less than optimal fit for Models 4 and 5. A final suggested path from social media pressure to body dissatisfaction was freed in Model 6. Freeing this path resulted in good fit across all indices. No further paths were suggested to be freed. All paths in the final model were significant at the $p < .05$ (see Figure 5). Although the χ^2 for the model was significant (Bollen Stine correction, $p = .006$), all other fit indices (including the normed χ^2) showed this to be a well-fitting model (see Figure 5).

4. Discussion

This study aimed to test two versions of the tripartite influence model (TIM: Thompson et al., 1999): one that included the original thin-ideal internalisation construct and one that included the more recent fit-ideal internalisation construct. Two minor modifications were made to the original TIM model for both versions tested: a path from thin/fit-ideal internalisation to social comparison was included, and compulsive exercise was included as an outcome variable.

All of the hypothesised paths were significant for both the thin- and fit-ideal internalisation models. Consistent with the TIM and hypotheses, the paths from family, peers and media pressures to thin / fit-ideal internalisation and social comparison were significant; the paths from thin / fit-ideal internalisation and social comparison to body dissatisfaction were significant; the paths from body dissatisfaction to compulsive exercise, dieting and bulimic symptoms were significant; and the paths from thin / fit-ideal internalisation to social comparison were significant. The final models for both thin- and fit-ideal internalisation included extra, non-hypothesised direct paths from media to body dissatisfaction, and from social comparison to dieting, bulimic symptoms and dieting. The two models differed in that the final model for thin-ideal internalisation included direct paths from thin-ideal internalisation to dieting and bulimic symptoms. In contrast, the final model for fit-ideal internalisation included a direct path from fit-ideal internalisation to compulsive exercise.

Turning first to the effects of sociocultural pressures, the results suggested significant paths from all three types of pressures (family, peers, and media) to thin-ideal internalisation, fit-ideal internalisation and social comparison. For the thin ideal model, this is in agreement with the vast amount of empirical research conducted to date on these pathways (Ata, Schaefer, & Thompson, 2015; Tiggemann, 2011). For fit-ideal internalisation however, these findings are novel and are in contrast to the findings of Ramme et al (2016) who found no association between any of the sociocultural pressures and ‘athletic’ (i.e., muscular) ideal

internalisation, highlighting (a) the importance of correct construct measurement and (b) that the 'fit' ideal is indeed different to the muscular ideal.

While all predicted paths from sociocultural pressures to thin- and fit-ideal internalisation and social comparison were significant, it is noteworthy that some of the coefficients were very small, suggesting that their significance may have been due to the large sample size. With respect to the thin ideal model, peer and family influences had weak (< .20) associations with thin-ideal internalisation, while the association between media influence and thin-ideal internalisation was stronger. In regard to the fit ideal model, family influences had a weak association with fit-ideal internalisation, while the associations between peer and media influences and fit-ideal internalisation were stronger. With respect to the relationships between sociocultural influences and social comparison, in both models the relationships between peer and family influences and social comparison were weak, while the relationship between media and social comparison was stronger. Interestingly, for both thin and fit ideal models, an unexpected finding was the direct (albeit weak) path from media influence to body dissatisfaction. It would seem that in current western society, family influences have less effect on the degree to which women take on societal body ideals and compare their bodies to other women, and that media are particularly influential with respect to internalisation of body ideals, social comparison, and body dissatisfaction.

The stronger associations between media influence on the one hand, and thin/fit internalisation, social comparison and body dissatisfaction on the other, may speak to the prolific use of social media and the use of thin and fit imagery within it. Although traditional media also promulgates fit and thin images, social media relies heavily on imagery, is constantly accessible, and is used more frequently (Instagram, 2020; Sensis, 2018). It is therefore likely to be particularly influential on body dissatisfaction, both directly and through internalisation of the ideals and social comparison. Indeed, social media is a form of

new age media, that can theoretically incorporate imagery from various sources including advertisements, celebrities, peers, and family. The fit ideal is widely propagated on social media (e.g., through fitspiration posts), yet the thin equivalent of this promotion (i.e., thinspiration) has been banned on platforms such as Instagram (Instagram, 2020). The result of this is an online culture that is more overtly endorsing of fitness than thinness (e.g., Raggatt et al., 2018). Given the prevalence of social media use amongst young women (Sensis, 2018), it is possible that the stronger influence of peers on fit- compared to thin-ideal internalisation in the current study, reflects the increased means by which young women can now compare themselves to their peers on this construct using such mediums.

With respect to the outcomes of thin- and fit-ideal internalisation, and as hypothesised, both constructs were related to body dissatisfaction. Again, although these effects have been repeatedly demonstrated for thin-ideal internalisation (Ata et al., 2015; Tiggemann, 2011), the results for fit-ideal internalisation are novel and somewhat contrary to what has previously been found. Indeed, it has often been concluded that 'athletic-ideal' internalisation does not work through body dissatisfaction to produce negative outcomes in the same way that thin-ideal internalisation does (e.g., Bell et al., 2016; Homan. 2010; Ramme et al., 2016). However, these studies utilised measures of muscular-ideal internalisation, which is a very different body standard to that of the fit ideal. The results of the present study suggest that when operationalised more accurately, there is good evidence for a relationship between fit-ideal internalisation and body dissatisfaction in women.

The additional direct pathways that were not predicted from thin-ideal internalisation to dieting and (albeit weakly) bulimic behaviours, from fit-ideal internalisation to compulsive exercise, and from social comparison, to dieting, bulimic behaviours and compulsive exercise, are also interesting. It would seem that when women internalise a body ideal and / or compare themselves with other women, they do not necessarily have to feel dissatisfied

with their bodies to engage in problematic eating and exercise behaviours. This is perhaps not surprising given the way that dieting has become so enmeshed with the thin ideal and exercise has become so enmeshed with the fit ideal. The reason for these more direct paths may at least in part be attributed to the nature of diet and exercise-based media. Many women believe that social media is a useful and reliable source of information on diet and exercise (Raggatt et al., 2018), and the vast majority of fitspiration and other posts contain reference to dieting and exercise behaviour (Boepple et al., 2016). While a large proportion of this media has been shown to be guilt inducing and objectifying (Boepple et al., 2016), some forms are 'inspirational' in nature (Boepple and Thompson, 2016). Indeed, women who engage with fitspiration media, report using the imagery for 'goals' with the intent being 'to look like her' by engaging in the prescribed dieting and exercise behaviour and comparing their own results as a means to measure 'success' (Raggatt et al., 2018). Thus, it is possible that some women do not have to feel negatively about their bodies to spur them to engage in problematic eating and exercise behaviour (although clearly some do). Rather, they have learned what they have to do through social media to attain the ideal, have internalised the ideal and compared themselves to others to see how they measure up, and then simply go on to engage in the behaviours espoused by social media to help them attain their goal. Such behaviour might therefore be operating from a need to 'belong' or 'conform' rather than from a place of dissatisfaction (Raggatt et al., 2018). Future research, particularly of a longitudinal nature, should address these suppositions so that we may better understand the impact of media on body dissatisfaction, and problematic eating and exercise behaviours.

The results of this study not only emphasise the differences between thin- and fit-ideal internalisation, they also emphasise the similarities. Despite the media, fitness and supplement industries espousing the 'healthy' nature of the fit ideal, the results of this study suggest otherwise. They suggest that as women desire and place value on attaining the fit

ideal, they compare themselves with other women, are dissatisfied with their bodies, and engage in compulsive exercise, dieting and bulimic behaviours. Extending on studies that have shown the ill-effects of exposure to the fit ideal (e.g., Prichard et al., 2017, Tiggemann & Zaccardo, 2015), the results of the current study support the notion that internalisation of the fit ideal has detrimental outcomes for women. Given that this is the first study to find associations between fit-ideal internalisation and negative outcomes using structural equation modelling with an assessment tool specifically designed to assess fit-ideal internalisation, the findings are particularly important.

4.1. Strengths, Limitations and Suggestions for Future Research

This study benefited from a number of strengths. It was the first study to test the TIM with a dedicated, psychometrically sound measure of fit-ideal internalisation and to compare it to the original TIM. Furthermore, the sample size was large, all measures were psychometrically sound, and the data were analysed using advanced statistical techniques to test the models in their entirety. Despite its strengths however, the study also had a number of limitations. It was cross-sectional in nature and therefore temporal and causal relationships cannot be assumed. Thus, it is critical that future research test this model longitudinally. The study also used a student sample that was largely Caucasian, and therefore ethnic diversity was low, and participants were relatively young with a low household income. Future research should not only employ a more heterogenous sample to ensure generalisability of results, but should also investigate whether or not representation of the fit ideal in social media is more ethnically diverse than traditional media representation, and whether women tend to 'seek out' fitspiration and images that are ethnically similar to themselves. A final limitation of the study relates to the use of the BIQ as a measure of body dissatisfaction. Although the BIQ was chosen due to its greater applicability to the fit ideal, it uses discrepancy scores, which have been criticised for low reliability (Cafri, Van Den Berg, &

Brannick, 2010). Despite the BIQ demonstrating good reliability in this study, the use of a discrepancy-type measure of body dissatisfaction may have influenced the results. Future research should therefore seek to develop new measures of body dissatisfaction that do not rely on discrepancy scores and yet are still applicable to different types of body ideals.

In addition to the suggestions for future research above, future studies should explore the particular social pressures, personality types and demographic characteristics that differentially predict internalisation of the different body ideals. Experimental work could also be undertaken to provide further evidence for the associations found in this study, and to better assess the temporal relationships hypothesised in these models. Finally, given the negative outcomes with which fit-ideal internalisation is associated, it is important that future research develops and tests strategies to counteract fit-ideal internalisation, in order to educate women on the unrealistic nature of the fit ideal, and the perils of attempting to pursue it.

4.2 Conclusions

The female fit body ideal is promoted by society as being the pinnacle of health and a 'better' alternative to the thin ideal. However, this study demonstrates that internalisation of the fit body ideal leads women to compare themselves with others, feel dissatisfied about their bodies, and to engage in unhealthy eating and exercise behaviours, just as internalisation of the thin ideal does. It would seem that society has replaced one unrealistic female body standard with an arguably more unrealistic one, and one that is leading to equally deleterious outcomes for women. Healthy exercise and healthy eating practices are good. Unhealthy exercise and unhealthy eating practices that are engaged in for the purpose of achieving a body that society deems 'ideal' is not. It is time to put the health back into health and fitness and to cease the relentless promotion and pursuit of yet another version of the body beautiful.

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